

SagaraMesh

Solar & Wave Powered Autonomous Floating Sea Rescue Mesh

A judge speech, teammate briefing, technical explanation, and complete Q&A; pack for presenting SagaraMesh confidently.

Core tagline

Connecting fishermen. Assessing danger. Saving lives. Protecting coastal communities.

1. Opening Speech Before Judges

Good morning respected judges.

Our project is SagaraMesh — a solar and wave-powered autonomous floating sea rescue mesh for fishermen safety.

The problem we are addressing is very real. Many fishermen travel into the sea where mobile networks are weak or completely unavailable. During storms, cyclones, boat failure, medical emergencies, or accidents, they may not be able to contact rescue teams on time. Even if rescue teams are ready, they often do not get the exact location or real-time situation quickly.

SagaraMesh solves this by creating a floating communication and rescue network over the sea. It uses smart buoys placed across fishing routes. These buoys are powered by solar energy and wave energy, so they can operate for long periods with low maintenance. When a fisherman presses an SOS button, the boat device sends the GPS location to the nearest buoy. The buoy then relays the message through other buoys until it reaches the shore command center.

But SagaraMesh is not just an SOS system. It also supports two-way communication, weather and cyclone alerts, Tamil/local-language warnings, boat tracking, international boundary warnings, and rescue acknowledgement. In the advanced version, if distress is detected, the buoy can deploy a drone, emergency payload, or underwater robot to assess the situation and report back to the command center before the rescue team reaches the location.

Another important feature is self-repositioning. If the buoy's anchor gets detached due to strong waves, collision, or cyclone conditions, the buoy detects drifting using GPS and anchor sensors, activates thrusters, and moves back to its preset GPS coordinate.

Existing systems like ISRO DAT-SG, EPIRB, PLB, AIS, and GMDSS already help with marine distress communication, but most of them are individual-device based, satellite-based, costly, or focused mainly on distress alerts. SagaraMesh is different because it acts as a shared renewable-powered floating communication and rescue infrastructure for many fishermen together.

Our MVP is feasible using ESP32 or Raspberry Pi Pico, LoRa modules, GPS, SOS button, solar panel, battery, waterproof enclosure, shore receiver, and dashboard. For demonstration, we can show SOS transmission, GPS tracking, mesh relay, dashboard alert, acknowledgement, simulated drone deployment, and simulated anchor drift correction.

SagaraMesh is useful, scalable, affordable, and highly relevant for Tamil Nadu and India's coastal communities. Our aim is simple: Connecting fishermen. Assessing danger. Saving lives. Protecting coastal communities.

Thank you.

2. One-Minute Version

Good morning judges. Our project is SagaraMesh, a solar and wave-powered autonomous floating sea rescue mesh for fishermen. Many fishermen go into areas where mobile networks do not work. During storms, accidents, boat failure, or medical emergencies, they cannot contact rescue teams quickly. SagaraMesh solves this by placing renewable-powered smart buoys across fishing routes. When a fisherman presses SOS, the boat device sends GPS location to the nearest buoy. The buoy relays the message through the mesh network to the shore command center. The system also sends weather alerts, Tamil/local-language warnings, rescue acknowledgements, boat tracking, and international boundary warnings. In the advanced version, the buoy can deploy a drone, emergency payload, or

underwater robot to assess the situation and can self-reposition if its anchor detaches. Unlike existing systems, SagaraMesh is shared, low-cost, renewable-powered floating safety infrastructure — not just an individual SOS device. Our goal is to turn the sea itself into a safety and rescue network.

3. Simple Explanation for Teammates

- SagaraMesh is a network of smart floating buoys in the sea.
- It helps fishermen communicate during emergencies when mobile networks are unavailable.
- It sends SOS, GPS location, weather alerts, Tamil/local-language warnings, rescue acknowledgement, boat tracking, and international boundary warnings.
- The advanced version adds drone/payload/ROV deployment, solar + wave power, and GPS self-repositioning if the anchor detaches.
- For hackathon, the full ocean system can be shown as an MVP demo with simulated advanced features.

4. Problem and Solution

Problem Statement	Fishermen often travel into sea zones where mobile networks are unavailable. During emergencies such as cyclones, boat failure, health issues, accidents, or drifting, they cannot reliably send distress messages or receive warnings. This delay in communication leads to delayed rescue, loss of life, and fear among coastal families.
Solution Statement	SagaraMesh creates a shared floating safety network using solar and wave-powered smart buoys. These buoys form a mesh communication layer between boats and shore stations, enabling SOS alerts, GPS tracking, weather warnings, rescue acknowledgement, and emergency assessment through drones or underwater robots.

5. How It Works

Step	Action
1	Fisherman faces emergency
2	SOS button is pressed or distress is detected
3	Boat device captures GPS location
4	Signal reaches nearest SagaraMesh buoy
5	Buoy relays message through mesh network
6	Shore command center receives alert
7	Drone / payload / underwater robot is deployed if needed
8	Situation report reaches command center
9	Rescue team is dispatched
10	Acknowledgement is sent back to fisherman

6. Technical and Physical Explanation

Part	Main components / role
Boat Device	SOS button, GPS module, LoRa/radio module, battery, indicator LED or small display, waterproof casing. Sends boat ID, GPS location, emergency type, time, and device status.
Floating Buoy	Solar panel, wave energy harvester, battery, charge controller, onboard computer, LoRa/radio, antenna, GPS, IMU/compass, anchor tension sensor, thrusters, LED beacon, optional satellite, optional drone/ROV/payload.
Shore Command Center	Receiver gateway, antenna, dashboard, map interface, alert notification system, rescue coordination, weather warning integration, family notification option.
Physical Design	Marine buoy body, waterproof electronics enclosure, corrosion-resistant materials, stable anchoring, solar panels on top, wave-energy mechanism, and protected thrusters for emergency repositioning.

7. Existing Solutions and Differentiation

Existing Solution	What it does	How SagaraMesh differs
ISRO DAT-SG	Satellite distress alerts, acknowledgements, weather alerts for fishermen.	SagaraMesh is shared floating mesh infrastructure with rescue assessment.
EPIRB	Satellite emergency beacon.	SagaraMesh reduces per-fisherman cost through community buoys and adds local mesh + rescue modules.
PLB	Personal emergency locator.	SagaraMesh supports whole fishing routes and communities.
AIS MOB	Man-overboard alert to nearby vessels.	SagaraMesh relays to shore through buoys and supports local-language alerts.
GMDSS	Global maritime distress communication for vessels.	SagaraMesh targets small coastal fishermen and local community deployment.
Tsunami / ocean buoys	Monitor ocean conditions.	SagaraMesh combines communication, warning, rescue assessment, and monitoring.

8. Strengths, Feasibility, Scalability, and Affordability

Strengths

- Life-saving emergency communication
- Works without mobile network
- Solar + wave powered
- Two-way communication
- Community-level safety network
- Drone/payload/ROV emergency assessment
- Self-repositioning buoy
- Tamil/local-language alerts
- Expandable for weather and environment monitoring

Feasible MVP

- Boat SOS transmitter
- GPS module
- LoRa communication
- Solar-powered buoy prototype
- Shore receiver
- Web dashboard
- Alert acknowledgement
- Simulated drone/payload deployment
- Simulated anchor drift correction

Scalability

- Prototype → Village pilot → District coastline → Tamil Nadu coastline → Indian coastal network

Affordability

- Low-cost electronics for MVP
- LoRa has low operating cost
- Solar/wave power reduces running cost
- One buoy supports many boats
- Fishermen need only a simple boat device
- Community/government/NGO funding reduces individual burden

9. Judge Q&A; Bank

Question	Answer
What exact problem are you solving?	We are solving unreliable emergency communication for fishermen at sea, especially where mobile networks are unavailable. The system helps send SOS, share location, receive warnings, and coordinate rescue quickly.
Why is this important?	Delayed communication during sea emergencies can cost lives. Fishermen face storms, cyclones, engine failure, medical emergencies, and navigation risks. Faster communication means faster rescue.
Why focus on fishermen?	Fishermen are highly vulnerable because they work in unpredictable sea conditions, often far from shore. Tamil Nadu and India have large coastal communities that depend on safe fishing livelihoods.
Is mobile network not enough?	No. Many fishing routes have weak or no mobile signal, and disaster conditions can damage communication infrastructure. SagaraMesh provides an independent sea communication layer.
What is SagaraMesh in one sentence?	SagaraMesh is a solar and wave-powered floating buoy mesh network that helps fishermen send SOS, receive alerts, and get rescue assessment support at sea.
How does SOS reach shore?	The boat device sends GPS and SOS data to the nearest buoy. The buoy forwards it through nearby buoys until it reaches the shore gateway and command center.
What if one buoy fails?	Mesh networking allows nearby buoys to route messages through alternate paths, reducing single-point failure.
Doesn't ISRO DAT-SG already exist?	Yes, DAT-SG is important. SagaraMesh differs because it is shared floating infrastructure using local buoy-to-buoy mesh, renewable power, and autonomous rescue assessment, not only an individual satellite distress device.
How are you different from EPIRB/PLB?	EPIRB and PLB are primarily individual emergency beacons. SagaraMesh is community infrastructure supporting many boats, two-way alerts, tracking, local-language warnings, drone/ROV assessment, and environmental extensions.
Is the idea completely new?	The components exist individually, but our novelty is combining them into a shared solar/wave-powered floating communication and autonomous rescue mesh for coastal fishermen.
What communication technology is used?	For MVP, LoRa is used because it is long-range, low-power, low-cost, and suitable for small data like SOS, GPS, and alerts. Advanced versions can add VHF, satellite backup, AIS integration, or cellular near shore.
Can LoRa send live video?	No. LoRa is for small messages. Drone video would need a separate higher-bandwidth link or compressed images/status reports.
How is the buoy powered?	It uses solar panels, wave energy harvesting, rechargeable batteries, and power management. Critical SOS communication is prioritized during low-power conditions.
How does anchor detachment detection work?	The buoy uses anchor tension sensing, GPS drift monitoring, IMU motion patterns, and distance from preset GPS coordinates. If drifting is detected, it alerts command center and activates thrusters.
Will thrusters consume too much power?	Thrusters consume more power, so they are used only during anchor detachment or emergency repositioning, not continuously.
Is underwater robot necessary?	No. It is an advanced module. The MVP focuses on communication, GPS, dashboard, and simulated deployment.
Can you actually build it?	Yes. The MVP is feasible using ESP32/Pico, LoRa, GPS, SOS button, solar panel, battery, waterproof box, receiver gateway, and dashboard. Advanced features can be modular upgrades.

Question	Answer
What is the hardest part?	Marine-grade durability, waterproofing, wave energy reliability, drone docking in rough sea, autonomous repositioning, permissions, and maintenance.
How will you test it?	Lab communication test, field LoRa range test, water-tank buoy test, pond/lake test, harbor pilot, then controlled sea pilot with authorities.
Is it affordable?	Yes in model: the buoy network is shared infrastructure. Fishermen need only a simple boat device, while government/cooperatives/NGOs/CSR can fund buoys.
Who benefits?	Fishermen, families, coastal villages, rescue teams, Coast Guard, disaster management authorities, fisheries department, and environmental monitoring agencies.
What are the limitations?	Real-sea deployment is difficult, hardware must be rugged, LoRa bandwidth is low, drone docking is hard in rough weather, permissions are needed, and maintenance is required.
What if the system fails?	It should not replace official safety systems. It acts as an additional safety layer alongside VHF, DAT-SG, EPIRB, PLB, and other systems.
Why can this win?	It solves a life-or-death problem, is relevant to Tamil Nadu/India, combines hardware, renewable energy, communication, robotics, and disaster management, has a feasible MVP, and scales from one village to a coastline.

10. Final Closing Statement

To conclude, SagaraMesh is not only about sending an SOS. It is about creating a safety layer over the sea.

- It connects fishermen when mobile networks fail.
- It sends warnings before danger reaches them.
- It gives rescue teams exact location and situation reports.
- It uses renewable power from the sun and waves.
- It supports communities through shared infrastructure.

SagaraMesh turns the sea itself into a safety and rescue network.

Thank you.